UNIT - I (COMPUTER – AN INTRODUCTION)

1.1 What is a Computer?

A **computer** is an electronic programmable machine that accepts data as input, processes it according to a set of instructions, stores the information, and produces meaningful results as output.

In simple words:

Computer is an electronic device that accepts data, processes it, stores it and produces useful information as output.

Example

Suppose a student enters:

```
Marks in 5 subjects:
80, 75, 90, 70, 85
```

The computer can:

- Accept the marks as **input**
- Calculate total and percentage as **processing**
- Store the marks as **data**
- Display total and percentage as **output**

For example:

```
Total = 400
Percentage = 80%
```

1.2 Data and Information

Students often get questions on **Data vs Information**, so this concept should be very clear.

Data

Data means raw and unprocessed facts and figures.

Examples:

```
85, Neeraj, BBA, 2026
```

These values alone may not provide complete meaning.

Information

Information is processed and meaningful data that can be used for decision-making.

Example:

If the marks are:

```
85, 90, 75, 80, 70
```

After processing:

```
Total Marks = 400
Percentage = 80%
```

This is information.

Difference between Data and Information

Data	Information
Raw facts	Processed facts
Usually unorganized	Organized and meaningful
May not be directly useful	Useful for decision-making
Input to a computer system	Output after processing
Example: 80, 90, 70	Example: Average = 80

Evolution of Computers (History)

The evolution of computers refers to the development of computing machines from simple calculating tools to the highly intelligent systems we use today.

The journey of computers began with basic mechanical devices and moved forward through various stages of technological advancement.

Early Calculating Devices

Device	Inventor	Year	Description
Abacus	Chinese Civilization	~3000 BC	First manual calculator using beads.
Napier's Bones	John Napier	1617	Used for multiplication and division.
Pascaline	Blaise Pascal	1642	First mechanical calculator.
Analytical Engine	Charles Babbage	1837	Considered the first concept of a modern computer.

Generations of Computers

The development of computer technology is divided into **five generations**, each marked by significant advancements in hardware and software.

1st Generation Computers (1940–1956)

- **Technology Used:** Vacuum Tubes
- **Programming Language:** Machine language (binary code – 0s and 1s)
- **Size:** Very large (occupied entire rooms)
- **Speed:** Very slow
- **Examples:** ENIAC, EDVAC, UNIVAC
- **Drawbacks:**
    - Produced a lot of heat
    - Required air conditioning
    - Frequently failed

2nd Generation Computers (1956–1963)

- **Technology Used:** Transistors (replaced vacuum tubes)
- **Programming Language:** Assembly language
- **Size:** Smaller and more reliable
- **Speed:** Faster than 1st generation
- **Examples:** IBM 1401, IBM 7094
- **Advantages:**
    - Less heat
    - Less power consumption
    - Better performance

3rd Generation Computers (1964–1971)

- **Technology Used:** Integrated Circuits (IC)
- **Programming Language:** High-level languages (like FORTRAN, COBOL)
- **Size:** More compact
- **Speed:** Much faster
- **Examples:** IBM System/360
- **Advantages:**
    - Lower cost
    - Better storage and processing
    - Supported multitasking

4th Generation Computers (1971–2010)

- **Technology Used:** Microprocessors (first was Intel 4004)
- **Programming Language:** All modern programming languages
- **Size:** Very small – personal computers
- **Speed:** Very high processing speed
- **Examples:** Desktop computers, laptops, tablets
- **Advantages:**
    - Affordable for personal use
    - Highly reliable
    - Use of GUI (Graphical User Interface), mouse, keyboard

5th Generation Computers (2010 & Beyond)

- **Technology Used:** Artificial Intelligence (AI), robotics, cloud computing, quantum computing
- **Programming Concepts:** Machine learning, Natural Language Processing (NLP)
- **Features:**
- Self-learning systems
- Voice and image recognition
- Real-time processing
- **Examples:** AI assistants (Alexa, Siri), smart robots, quantum computers
- **Advantages:**

  - Intelligent decision making
  - Automation of complex tasks
  - IoT (Internet of Things) integration

Types Of Computer

a) Analog Computer:

- Works on continuous data (not digital).
- Used in scientific and engineering applications.
- Example: Speedometer, thermometer.

b) Digital Computer:

- Works on discrete data (0 and 1 – binary).
- Most common type of computer used today.
- Example: Laptop, mobile phone.

c) Hybrid Computer:

- Combination of analog and digital computer.
- Used in hospitals (e.g., ECG machines), scientific applications

Types Of Digital Computer

a) Microcomputer (Personal Computer – PC):

- Smallest and most commonly used.
- Designed for individual use.
- Example: Desktop, Laptop.

b) Mini Computer:

- Mid-sized computer.
- Supports multiple users at the same time.
- Used in small and medium organizations.

c) Mainframe Computer:

- Large and powerful systems.
- Can support hundreds or thousands of users simultaneously.
- Used in banks, railways, government departments.

d) Supercomputer:

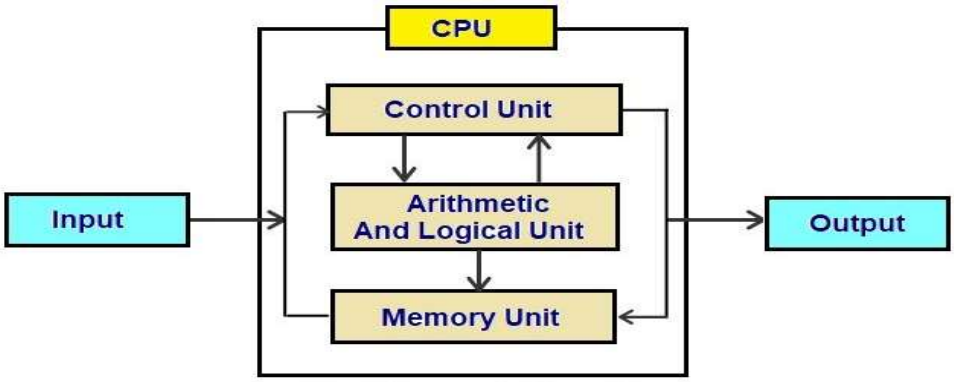
- Most powerful and fastest computer.
- Used in scientific research, weather forecasting, space research.
- Example: PARAM, Fugaku.

1.3 Basic Functions of a Computer

A computer generally performs five basic functions:

1. Input
2. Processing
3. Output
4. Storage
5. Control

Basic Computer Working



The control unit coordinates these activities.

1.3.1 Input

- **Input Devices** are hardware components that are used to **enter data and instructions** into the computer system for processing.
- These devices allow the user to interact with the computer by providing raw data, which is later processed by the CPU.

Functions of Input Devices:

- Accept data from the user.
- Convert human-readable data into machine-readable form (binary).
- Send data and instructions to the CPU for processing.

Examples of input devices:

- Keyboard, Mouse
- Scanner, Microphone
- Webcam, Touchscreen
- Barcode Reader

1.3.2 Processing

Processing means performing operations on input data to convert it into useful information.

Processing is mainly performed by the **CPU – Central Processing Unit**.

1.3.3 Output

- **Output Devices** are hardware components used to **display, print, or present the results** of data processing performed by the computer.
- They convert **machine-readable data** (binary) into **human-readable form** such as text, images, sound, or printed output.

Functions of Output Devices:

- Receive processed data from the CPU.
- Convert it into meaningful output.
- Present the result to the user in different forms (visual, audio, printed).
- Types of Output Devices:

Examples:

- Monitor, Printer
- Speaker, Projector

1.3.4 Storage

A computer stores data, programs and information for future use.

Examples:

- Hard Disk, SSD, Pen Drive, Memory Card
- CD/DVD, Cloud Storage

Storage can broadly be classified as:

Primary Storage

Examples:

- RAM
- ROM
- Cache Memory

Secondary Storage

Examples:

- HDD
- SSD
- Pen Drive
- Memory Card

1.3.5 Control

The computer must coordinate all operations.

The **Control Unit (CU)** of CPU controls and coordinates different activities of the computer.

Main Components of CPU

The CPU is mainly divided into **three parts**:

ALU (Arithmetic and Logic Unit)

- Performs all **arithmetic operations** like addition, subtraction, multiplication, division.
- Performs **logical operations** like comparison (greater than, less than, equal to).
- It is the **calculating unit** of the CPU.

CU (Control Unit)

- Manages and controls all activities inside the computer.
- It directs the flow of data between memory, ALU, and input/output devices.
- Does **not process** data itself but **controls** how other parts work.

Registers

- Small high-speed memory inside the CPU.
- Temporarily stores data, instructions, or results during processing.
- Faster than RAM and helps in **quick access** to frequently used data.

How CPU Works (In Simple Steps):

- **Fetch** – The control unit fetches instructions from memory.
- **Decode** – The instruction is decoded to understand the operation.
- **Execute** – ALU executes the instruction.
- **Store** – Result is stored in memory or sent to output.

Important Points:

- CPU is installed on the **motherboard**.
- The performance of a computer depends largely on the **speed and power of the CPU**.
- CPU speed is measured in **GHz (Gigahertz)**.
- Modern CPUs are **multi-core**, meaning multiple processing units in one chip (like Dual-core, Quad-core).

1.4 Characteristics of Computers

Computers have several important characteristics.

1. Speed

Computers can perform millions or billions of operations in a very short time.

For example, calculations that may take humans several minutes can be performed by a computer in fractions of a second.

2. Accuracy

Computers generally provide highly accurate results when:

- Data is correct
- Instructions/program are correct

This is often described as:

Garbage In, Garbage Out (GIGO)

If incorrect data is given to the computer, the output may also be incorrect.

Example:

```
Salary = ₹20,000
```

If the user mistakenly enters:

```
Salary = ₹2,00,000
```

the computer will calculate based on the incorrect input.

3. Storage Capacity

Computers can store large amounts of data.

For example, an organization can store:

- Employee records
- Customer information
- Sales records
- Financial records
- Inventory details

4. Diligence

A computer does not become tired or bored like humans.

It can perform the same operation repeatedly with consistent performance.

Example:

A computer can process thousands of salary records without getting tired.

5. Automation

Once a program and required data are provided, a computer can perform tasks automatically.

Example:

A bank's system can automatically calculate:

- Interest
- Account balance
- Transaction records

6. Versatility

Computers can be used for many different purposes.

For example:

- Education
- Banking
- Business
- Healthcare
- Entertainment
- Communication
- Manufacturing
- Government services

7. Reliability

Computers can perform repetitive operations consistently.

However, reliability depends on:

- Hardware
- Software
- Correct input
- Proper maintenance

8. Multitasking

Modern computers can perform multiple tasks apparently at the same time.

For example, a user may:

- Play music
- Browse the Internet
- Download a file
- Edit a document

at the same time.

1.5 Limitations of Computers

Although computers are powerful, they also have limitations.

1. No Intelligence by Default

A computer does not independently understand a problem like a human unless appropriate software or AI systems are designed for it.

2. No Common Sense

Computers follow instructions and rules.

3. Dependency on Instructions

A computer needs instructions/programs to perform tasks.

4. Incorrect Input Produces Incorrect Results

This is known as:

GIGO – Garbage In, Garbage Out

5. No Emotions

Computers do not have human emotions or feelings.

6. Maintenance Requirement

Hardware and software require maintenance and updates.

1.6 Applications of Computers

Computers are used in almost every field.

Education

Computers are used for:

- Online classes
- Digital learning
- Examination systems
- Student records
- Presentations
- Research

Banking

Computers are used for:

- ATM
- Online banking
- Account management
- Fund transfer
- Transaction processing

Business

Businesses use computers for:

- Accounting
- Payroll
- Inventory
- Sales
- Customer management
- Business analysis

Healthcare

Computers are used for:

- Patient records
- Medical imaging
- Hospital management
- Laboratory systems
- Appointment management

Government

Computers are used for:

- Digital records
- Tax management
- Online services
- E-governance
- Public databases

Communication

Computers support:

- Email
- Video conferencing
- Messaging
- Social media
- Internet communication

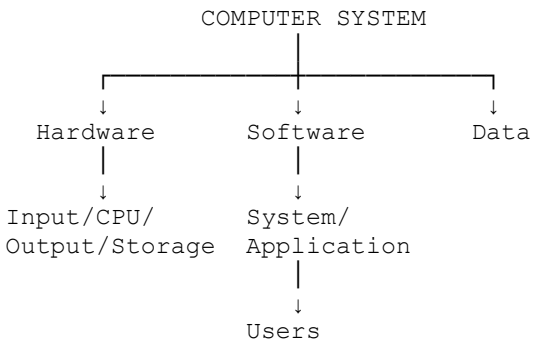
2. COMPONENTS OF COMPUTER SYSTEM

A computer system is not just a CPU or monitor. It consists of different components that work together.

The major components are:

1. Hardware
2. Software
3. Data
4. Users
5. Procedures

A simple representation is:



2.1 Hardware

Hardware refers to the physical parts of a computer system that can be seen and touched.

Examples:

- Keyboard, Mouse, Monitor, CPU
- Motherboard, RAM, Hard Disk, SSD
- Printer, Scanner, Speakers

Major Hardware Categories

Hardware can be broadly divided into:

1. Input devices
2. Processing devices
3. Output devices
4. Storage devices
5. Communication devices

2.2 Input Devices

Input devices are used to enter data and instructions into the computer.

Keyboard

A keyboard is an input device used to enter:

- Characters
- Numbers
- Symbols

- Commands

Mouse

A mouse is a pointing device.

Common mouse operations:

- Click
- Double-click
- Right-click
- Drag and drop
- Scroll

Scanner

A scanner converts physical documents or images into digital form.

Example:

A printed document can be scanned and stored as a digital file.

Microphone

A microphone is used to input sound/audio into a computer.

Webcam

A webcam captures images and video.

Applications include:

- Video conferencing
- Online classes
- Video calls

Barcode Reader

A barcode reader reads information stored in barcodes.

Commonly used in:

- Retail stores
- Supermarkets
- Inventory systems

QR Code Scanner

QR codes can store information such as:

- Website URLs
- Payment information

- Contact information
- Product information

2.3 Central Processing Unit – CPU

The **CPU (Central Processing Unit)** is responsible for executing instructions and performing processing operations.

It is often called the **brain of the computer**, although technically it is the main processing unit rather than a human-like brain.

The CPU mainly consists of:

1. **ALU – Arithmetic Logic Unit**
2. **CU – Control Unit**
3. **Registers**

Modern CPUs also contain additional components such as cache memory and multiple processing cores.

2.4 ALU – Arithmetic Logic Unit

ALU performs:

Arithmetic Operations

- Addition
- Subtraction
- Multiplication
- Division

Logical Operations

- Greater than
- Less than
- Equal to
- AND, OR, NOT

Example:

```
10 + 20 = 30
```

This calculation is handled by the processing unit, with arithmetic operations performed by the ALU.

2.5 Control Unit – CU

The **Control Unit** controls and coordinates the operations of the computer.

Its major responsibilities include:

- Fetching instructions

- Decoding instructions
- Controlling execution
- Coordinating CPU components
- Managing movement of data and instructions

The basic instruction cycle is:

```
Fetch → Decode → Execute
```

2.6 Registers

Registers are very small and very fast storage locations inside the CPU.

They temporarily hold:

- Data
- Instructions
- Addresses
- Intermediate results

Examples include:

- Accumulator
- Program Counter
- Instruction Register
- Memory Address Register

2.7 Memory

Memory is used to store:

- Data
- Instructions
- Programs
- Intermediate results

Memory is generally divided into:

Primary Memory

- RAM
- ROM
- Cache

Secondary Memory

- HDD
- SSD
- Pen Drive
- Memory Card
- Optical disks

2.8 Output Devices

Output devices present processed information to the user.

Examples:

- Monitor
- Printer
- Speaker
- Projector

Monitor

A monitor displays:

- Text
- Images
- Videos
- Graphics

Printer

A printer produces a physical copy of digital information.

This physical copy is called a **hard copy**.

Types include:

- Inkjet printer
- Laser printer
- Dot matrix printer

Speaker

A speaker produces audio output.

Projector

A projector displays computer output on a large screen.

It is commonly used in:

- Classrooms
- Meetings
- Seminars
- Conferences

2.9 Software

Software is a collection of programs and instructions that tells the computer hardware what to do.

Unlike hardware, software cannot normally be physically touched.

Examples:

- Windows
- Linux
- MS Word
- MS Excel
- Google Chrome
- VLC Media Player

Software can broadly be divided into:

1. System Software
2. Application Software

2.10 System Software

System software manages computer hardware and provides a platform for application software.

Examples:

- Operating Systems
- Device Drivers
- Utility Programs
- Language Translators

Examples of operating systems:

- Windows
- Linux
- macOS
- Android

2.11 Application Software

Application software is designed to perform specific tasks for users.

Examples:

Software	Purpose
MS Word	Document processing
MS Excel	Spreadsheet and calculations
PowerPoint	Presentation
Chrome	Web browsing
Photoshop	Image editing
Tally	Accounting

Application software can further be:

General Purpose Software

Used by many types of users.

Examples:

- Word processors
- Spreadsheets
- Presentation software

Special Purpose Software

Designed for a particular purpose.

Examples:

- Railway reservation system
- Banking software
- Hospital management system
- Payroll system

2.12 Users / Peopleware

People who use or operate computer systems are sometimes collectively referred to as **peopleware**.

Examples:

- Computer operators
- Programmers
- System analysts
- Database administrators
- Managers
- End users

2.13 Procedures

Procedures are the rules or steps that explain how a computer system should be used.

For example, in an organization:

```
Login
↓
Enter employee details
↓
Verify information
↓
Save record
↓
Generate report
```

2.14 Complete Computer System

Therefore, a computer-based system can be understood as:

```
Hardware
   +
Software
   +
Data
   +
People
   +
Procedures
   =
Computer-Based Information System
```

Advantages of Computer System

1. **High Speed** – Performs calculations and tasks very quickly.
2. **Accuracy** – Produces accurate results when given correct instructions.
3. **Storage** – Can store a large amount of data.
4. **Automation** – Performs tasks automatically.
5. **Multitasking** – Can perform multiple tasks at the same time.
6. **Reliability** – Can work continuously without getting tired.
7. **Easy Communication** – Helps in fast communication through email, internet, etc.

Disadvantages of Computer System

1. **Security Risks** – Data can be stolen or hacked.
2. **Health Problems** – Excessive use can cause eye strain and other health issues.
3. **Dependency** – Excessive dependence on computers can reduce human effort and skills.
4. **Job Displacement** – Automation can replace some human jobs.
5. **Cost** – Hardware, software, maintenance, and upgrades can be expensive.
6. **Data Loss** – Data may be lost due to hardware failure, viruses, or accidental deletion.
7. **Cybercrime** – Computers can be affected by viruses, malware, and other cyber attacks.

3. INDIAN COMPUTING ENVIRONMENT

The **Indian Computing Environment** refers to the development, adoption and use of computer technology, information technology and digital systems in India.

India has developed a large computing ecosystem involving:

- Government
- Businesses
- Educational institutions
- Banks
- IT companies
- Startups
- Research organizations
- Citizens

3.1 Development of Computing in India

The development of computing in India can be broadly understood in stages:

Early Stage (1940s to 1960s)

Computers were initially used mainly by:

- Government organizations
- Scientific institutions
- Research organizations
- Large businesses

Expansion Stage (1970s to 1980s)

With decreasing hardware costs and increasing availability of technology, computers became common in:

- Offices
- Schools
- Colleges
- Banks
- Businesses

Internet Era (1990s to 2000s)

The growth of the Internet introduced:

- Email
- Websites
- E-commerce
- Online banking
- Digital communication

Digital Era (2010s to Present)

Modern India increasingly uses:

- Smartphones
- Cloud computing
- Digital payments
- Online government services
- E-commerce
- Artificial intelligence
- Data analytics
- Digital education

3.2 Role of Computers in India

Computers play an important role in:

Government

- E-governance
- Digital records

- Online applications
- Tax systems
- Public services

Banking

- ATM
- Internet banking
- Mobile banking
- Online transactions
- Digital payments

Education

- Online classes
- Digital libraries
- Learning management systems
- Online examinations

Business

- Accounting
- Marketing
- Customer management
- Inventory
- E-commerce

Healthcare

- Digital patient records
- Hospital management
- Medical research
- Telemedicine

3.3 Digitalization in India

Digitalization means using digital technologies to improve services and processes.

Examples include:

- Online banking
- UPI-based payments
- Online ticket booking
- Digital government services
- Online education
- E-commerce

Period	Stage	Major Development
1950–1970	Early Computing	Mainframes, TIFRAC, IBM computers
1970–1985	Computing Expansion	Business & government computerization

Period	Stage	Major Development
1985–1995	PC & IT Era	PCs, software, computer education, ERNET
1995–2005	Internet Expansion	ISP, cyber café, email, websites, VSNL public Internet
2005–2010	Broadband Era	Broadband, online services,Cable, Fibre Optics
2010–2016	Mobile Internet	3G, smartphones, apps
2015 onwards	Digital India	E-governance, digital services, e-Hospital
2016 onwards	Mass Internet	Jio, cheap mobile data, smartphones
2020 onwards	Digital Transformation	UPI, cloud, online education, remote work, digital businesses
2022 onwards	AI & Emerging Tech	AI, GenAI, cloud, IoT, 5G

3.4 Advantages of Computing Environment in India

1. Faster communication
2. Digital financial transactions
3. Better access to information
4. Improved government services
5. Growth of IT industry
6. Employment opportunities
7. Online education
8. Business automation
9. Better data management
10. Global connectivity

3.5 Challenges

India's computing environment also faces challenges such as:

- Cybersecurity threats
- Data privacy concerns
- Digital literacy gaps
- Unequal Internet access
- Infrastructure limitations
- Cyber fraud
- Need for skilled professionals

4. COMPUTERS IN BUSINESS

Computers have become an essential part of modern business organizations.

Businesses use computers to:

- Store information
- Process transactions

- Communicate
- Manage employees
- Maintain accounts
- Manage inventory
- Analyze sales
- Make decisions
- Communicate with customers

4.1 Major Applications of Computers in Business

1. Accounting

Computers are used for:

- Recording transactions
- Preparing invoices
- Calculating taxes
- Preparing financial statements
- Managing expenses

Example:

Accounting software can automatically calculate totals and generate reports.

2. Payroll Management

Businesses use computer systems to calculate:

- Employee salary
- Attendance
- Overtime
- Bonuses
- Deductions
- Taxes

3. Inventory Management

Computerized inventory systems track:

- Stock available
- Purchases
- Sales
- Stock movement
- Reorder levels

Example:

If inventory falls below a predefined level, the system can generate an alert.

4. Sales Management

Computers help businesses maintain:

- Sales records
- Customer information
- Product information
- Invoices
- Sales reports

5. Marketing

Computers are used for:

- Digital marketing
- Customer analysis
- Email marketing
- Social media marketing
- Advertising
- Market research

6. Human Resource Management

HR departments use computer systems for:

- Employee records
- Recruitment
- Attendance
- Payroll
- Leave management
- Performance records

7. Customer Relationship Management

CRM systems help businesses manage customer information and interactions.

They can store:

- Customer names
- Contact details
- Purchases
- Complaints
- Communication history

8. Decision Making

Managers can use computer-generated reports to make business decisions.

For example:

```
Sales Report
     ↓
Data Analysis
     ↓
Identify Best-Selling Product
     ↓
Business Decision
```

4.2 Advantages of Computers in Business

Speed

Large amounts of business data can be processed quickly.

Accuracy

Computerized calculations reduce manual calculation errors.

Cost Reduction

Automation can reduce repetitive manual work.

Better Record Keeping

Data can be stored and retrieved easily.

Better Communication

Email, messaging and video conferencing improve communication.

Better Decision Making

Reports and data analysis provide useful information to managers.

Automation

Routine tasks can be automated.